

CS 214 PROJECT





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INTRODUCTION



The Task Management System is a specialized C++ application designed to integrate advanced data structures with object oriented programming to achieve high performance and data persistence.

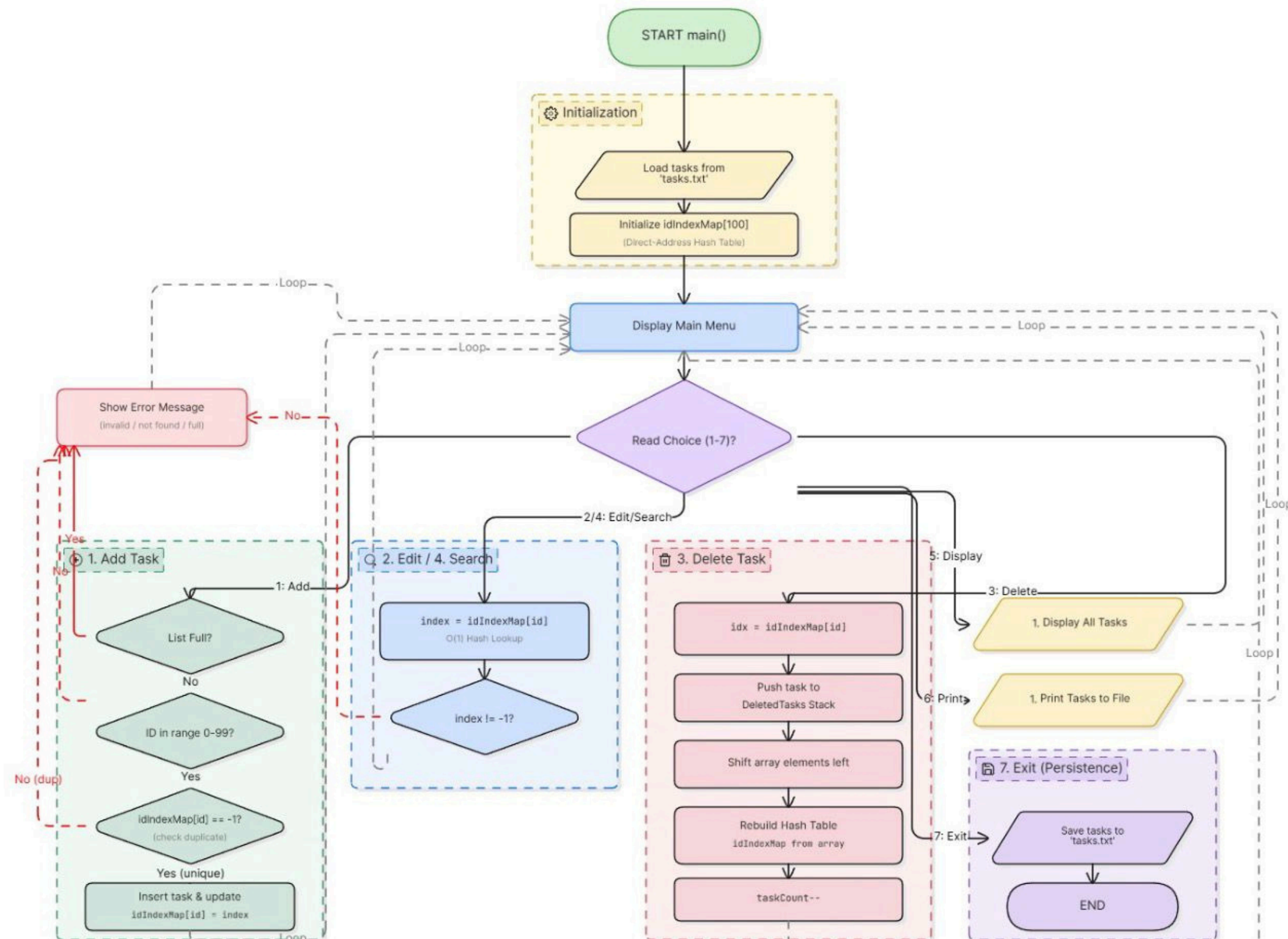
At its core the system utilizes a custom struct to encapsulate multiple data types ensuring that

every task is treated as a single coherent entity in memory



flow chart

C++ Task Management System - Flowchart



DATA STRUCT

Array

It stores all the tasks in a simple, side-by-side list. It is very efficient for saving a fixed amount of data in memory

Stack

delete a task, we put it in a Stack. Since a Stack follows the "Last-In, First-Out" rule, it is perfect for keeping track of deleted items

Hash Table

Searching one by one is slow. We used an Index Map as a "shortcut." It tells the program exactly where the task is located based on its ID

OUTPUT

Delete Function test output

```
==== Task Management System ====
. Add Task
. Edit Task
. Delete Task
. Search Task
. Display All Tasks
. Print Tasks to File
. Exit
Enter your choice: 5

All Tasks:
-----
D: 1
Title: Study math
Description: Solve homework
Priority: 1
-----
D: 2
Title: Read book
Description: Read 10 pages
Priority: 2
-----
D: 3
Title: Meet friend
Description: Go to cafe
Priority: 3
-----
D: 4
Title: Prepare for exam
Description: Review notes
Priority: 1
-----
D: 5
Title: Clean desk
Description: Organize papers
Priority: 3
-----

==== Task Management System ====
. Add Task
. Edit Task
. Delete Task
. Search Task
. Display All Tasks
. Print Tasks to File
. Exit
Enter your choice: 3
Enter task ID to delete: 5
Task deleted successfully.
```

**THANK
YOU**

